

~~Point Operation~~ Introduction

GMP

Image enhancement refers to the sharpening or to improve the quality of an image by enhancement of pixel or point in a spatial domain.

Some basic gray level transformation is needed to enhance the image in a spatial domain. Some basic gray level transformation is listed as below:

- (i) Point Operations
- (ii) Contrast Stretching
- (iii) Thresholding
- (iv) Digital Negative
- (v) Intensity Level Slicing
- (vi) Bit plane Slicing

Point Operation:

Point operation is defined as a function that are performed on each pixel of an image.

Point operation is unary point operation if single image is modified. The operation can be binary which means that two images combine in the same manner.

the operation is called as ternary if three images are used in same manner at operation.

Unary operation can be defined as,

$$T_{xy} = \text{function}(P(x, y))$$

where,

T_{xy} : Transformed Pixel Value

function : Numeric transform

$P(x, y)$: Input Pixel Value

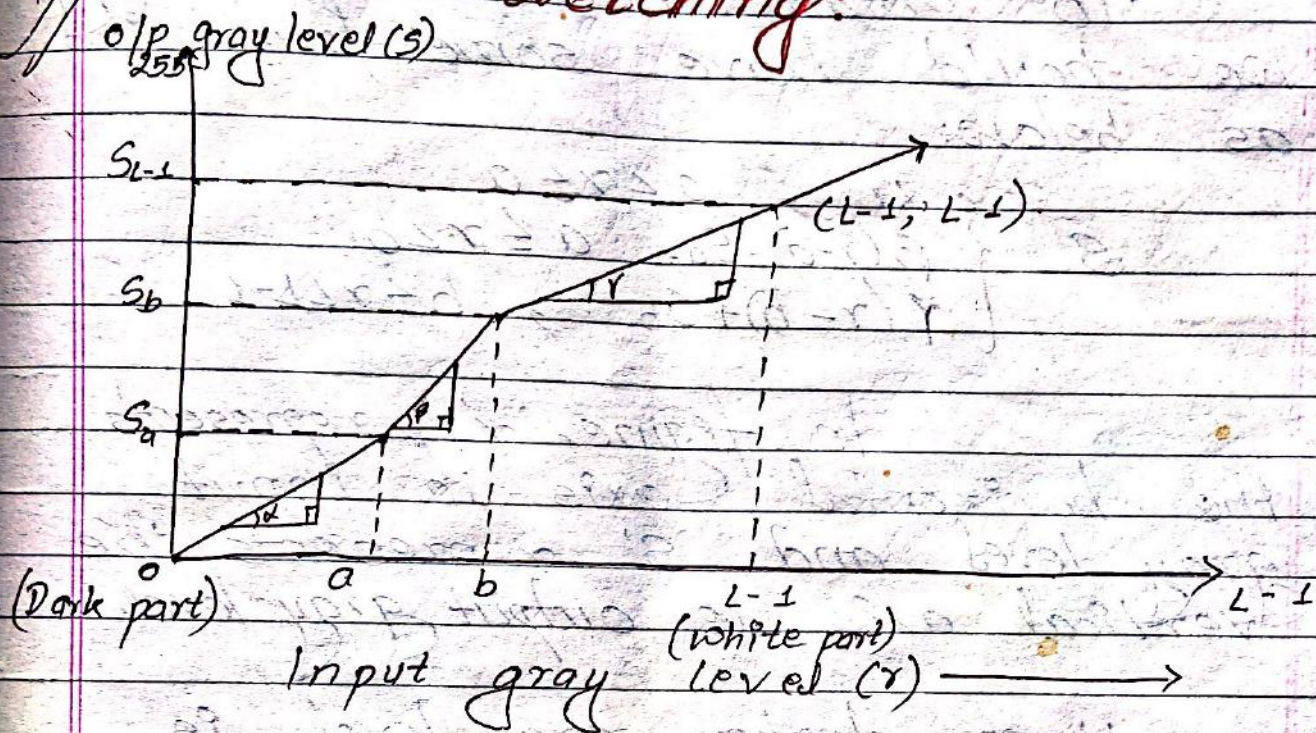
This operation can be used in many situations and different purpose such as: contrast stretching, noise clipping, window slicing and histogram modeling.

So, in point operation, a given gray level, $u \in [0, L-1]$ is mapped into a gray level $v \in [0, L-1]$ according to a transformation.

$$\text{i.e. } \boxed{v = f(u)}$$

$$\begin{cases} L-1 = 255 \\ L = \text{max gray level} \end{cases}$$

Contrast Stretching:



(fig: Contrast stretching transformation) where, α , β & γ are the slopes.

In many times, we obtain low contrast image due to poor illumination or due to the wrong setting of lens aperture.

The idea behind contrast stretching is to increase the dynamic range of gray level in the image. The main idea is to increase the contrast of the image by making the dark portion darker & bright portion brighter.

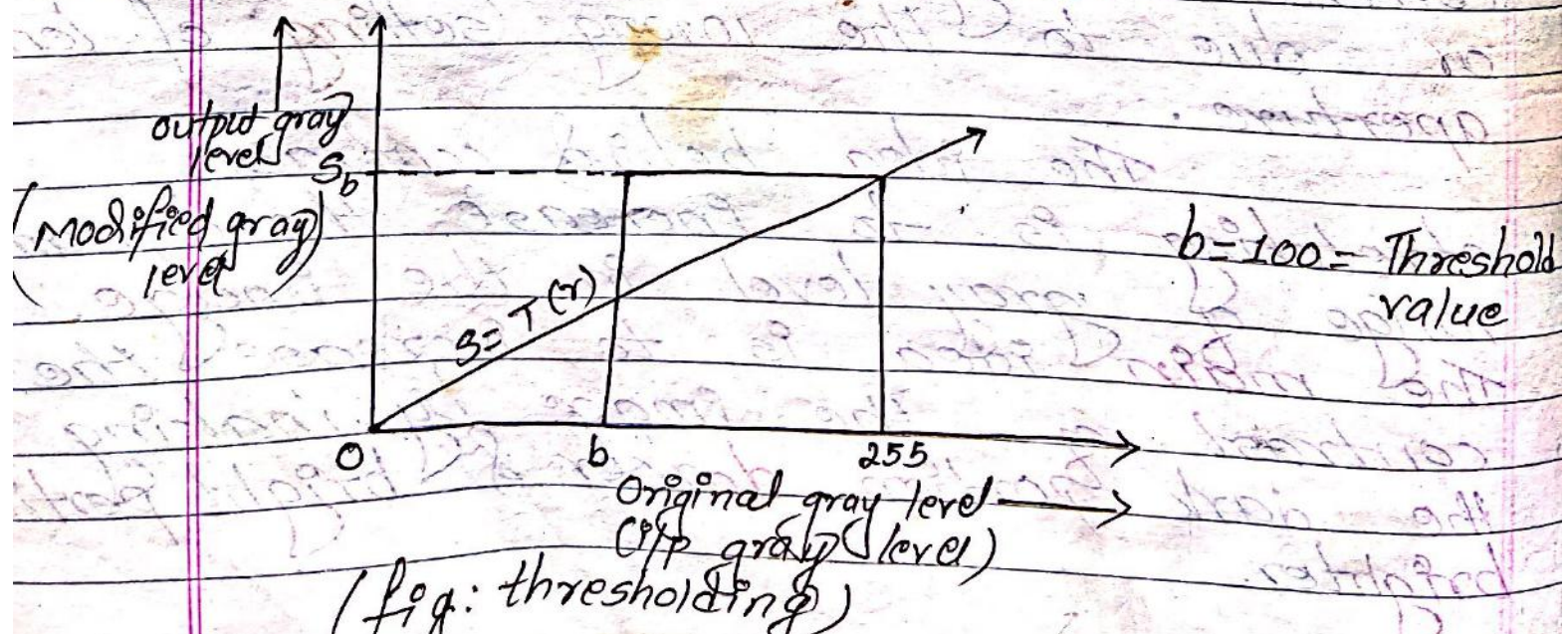
To obtain the contrast stretching we should have some formulations as below:

$$S = \begin{cases} \alpha r, & 0 \leq r \leq a \\ \beta(r-a) + S_a, & a \leq r \leq b \\ \gamma(r-b) + S_b, & b \leq r \leq L-1 \end{cases}$$

In figure, 'r' represents the horizontal axis i.e. input gray level and 'S' represents the vertical axis i.e. output gray level.

Here, darker gray level is obtained by making a slope < 1 and brighter is obtained by making a slope > 1 .

Thresholding (Binary Image)



It is the simplest method of the image processing that can be used to create binary image.

If we observe the contrast stretching diagram, we notice that the first and last slopes are made zero.

If $r_1 = r_2$ & $s_1 = 0$, $s_2 = L-1$

Now, the ~~corresponding~~ thresholding function is given by:

$$s = 0, \text{ if } r < b \rightarrow 0$$

$$s = L-1, \text{ if } r \geq b \rightarrow 1$$

where,

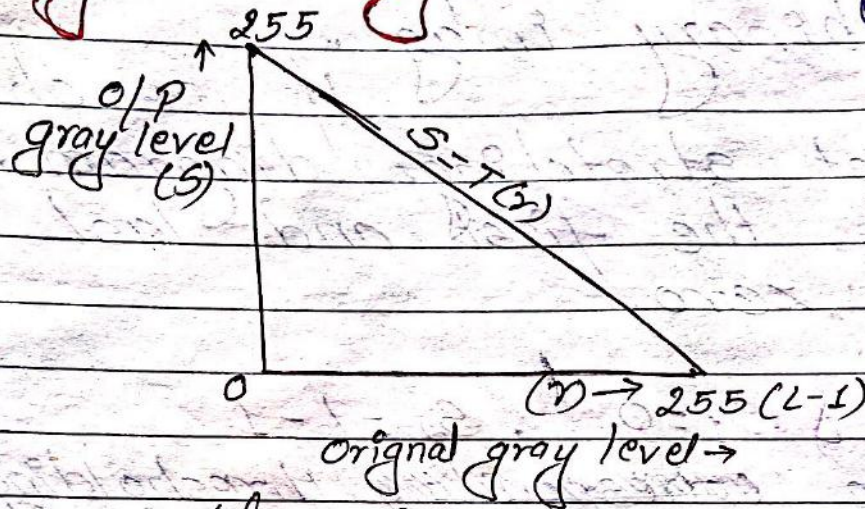
b - thresholding value

L - maximum gray level

Important notice is that, threshold image has maximum contrast on it having black and white gray level only.

$$T(r) = \begin{cases} 0 & \text{if } r < b \\ L-1 & \text{if } r \geq b \end{cases}$$

Digital Negative (X-Ray)



(fig: Digital Negative)

It is useful in a lots of application. Common example of digital negative is X-Ray image.

As the name suggest, negative means simply inverting the gray level ie, Black into white and vice versa.

So, the negative of digital image with gray level in the range from 0 to $(L-1)$ is obtained by using negative transformation.

$$\text{i.e., } S = T(r) = [(L-1) - r]$$

where,

r = input pixel value
 L = No. of gray level / Max. gray level

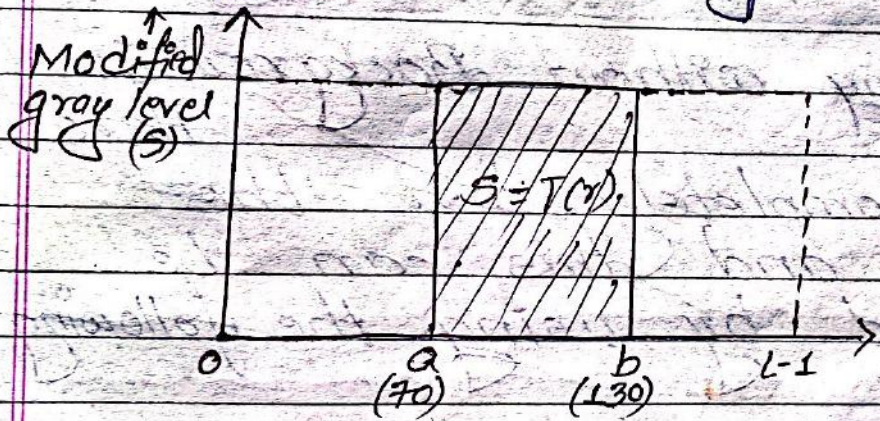
$T(r)$ = Transformation function
 S = output gray level

Hence,
 when $r = 0$, $S = 255$ (white)
 $r = 255$, $S = 0$ (black)

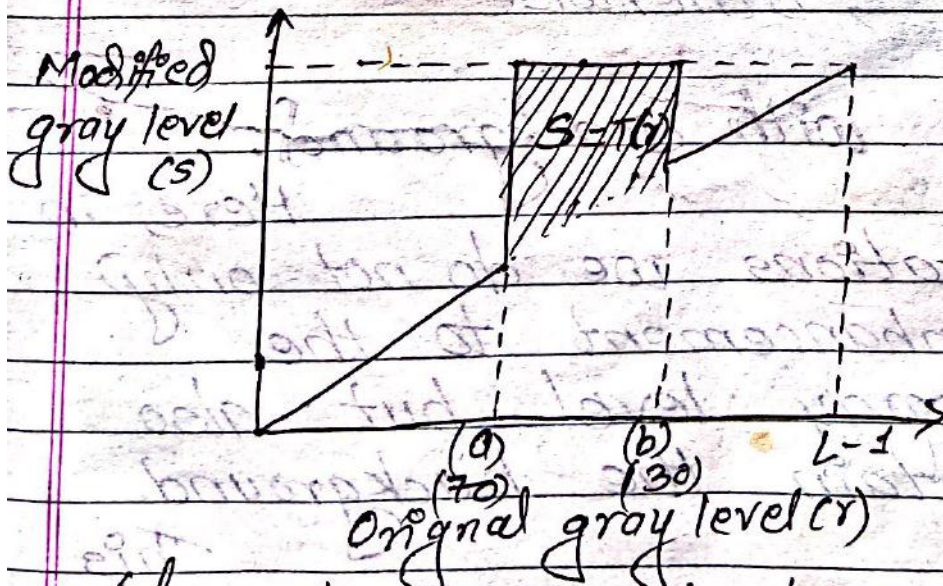
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Intensity level slicing

[Gray Level Slicing]



Original gray level (r) $\rightarrow$
 (fig: slicing without background)



Original gray level (r)
 (fig: slicing with background)

Thresholding splits the gray level into two parts. But in many times we need highlight a specific range of gray level.

In such situations, use transformation function on a gray level called as gray level ~~slit~~ slicing.

There are two types of slicing:

(i) Slicing without Background:

Here we have completely lost the background and this can be implemented by using the following function:

ie,

$$s = L - 1, \text{ if } a \leq r \leq b$$

$$s = 0, \text{ otherwise}$$

(ii) Slicing with Background:

Here, in some applications we do not only need of enhancement to the band of gray level but also need to retain the background.

This can be implemented by using

the following function:

$$\text{i.e. } S = L - 1, \text{ if } a \leq r \leq b$$

$$S = r, \text{ otherwise}$$

gmp.

Bit plane Slicing (Clipping)

It is used to enhance features with an image. We must provide a threshold level to determine how the clipping occurs.

The ~~threshold~~ threshold level or value is a reference value where other values get changed.

In general it is performed for maximum and minimum operators. An expression that contain an image with threshold levels.

For example: if ~~we~~ we have image variable & want to slice it to ≥ 125 , then the resulting image is stated as:

$$\text{Clipped Image} = (\text{Image(Original)}) > 125$$

Here, it means if the element values is less than

125, then it is set as equal to 125 otherwise it is unchanged.

V. Imp
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Introduction to Histogram

Histogram of an image provides a global description of the appearance of an image.

The information obtained from histogram is innermost & hence, histogram modeling is important in digital image processing.

By (the definition histogram of an image represents the relative frequency of occurrence of various gray level in an image.)

Histogram of an image can be plotted as:

- (i) The X-axis has the gray level and Y-axis has the number of pixels in each gray level.
- (ii) The X-axis represent gray level & Y-axis represents the probability of occurrence of that gray level.

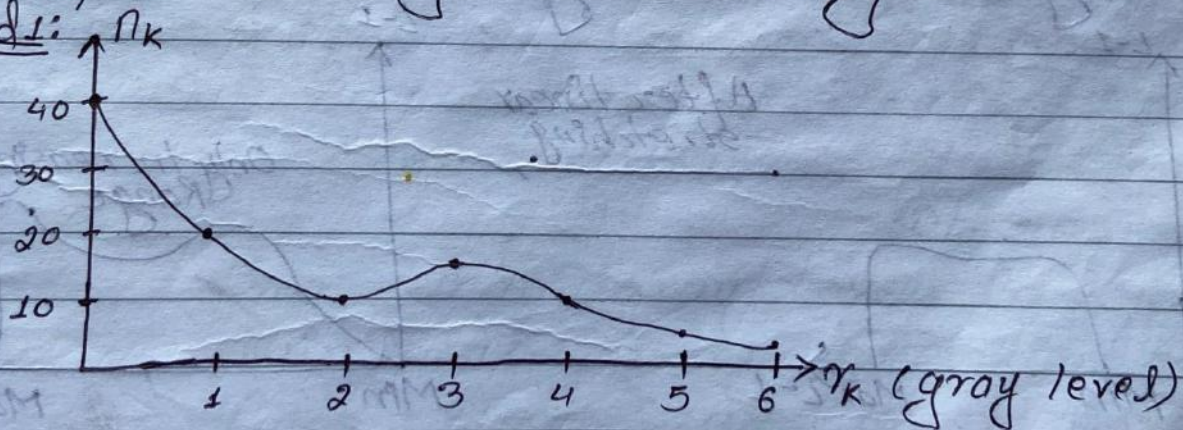
The method of histogram is shown as below:

Step 1: Original Image

Gray level (r_k)	No. of pixel (n_k)
0	40
1	20
2	10
3	15
4	10
5	3
6	2

Step 2: Histogram Modeling

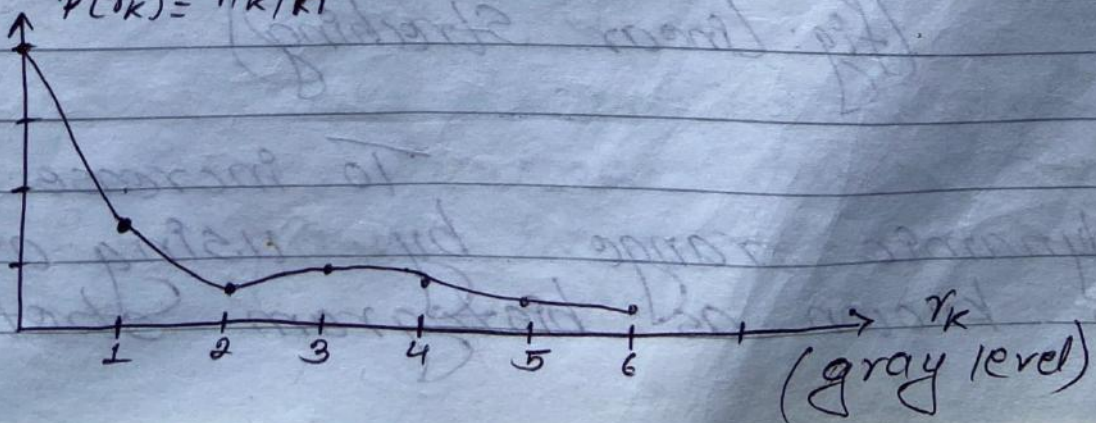
Method 1:



Method 2:

$$P(r_k) = n_k / N$$

Probability
of occurrence
of gray level

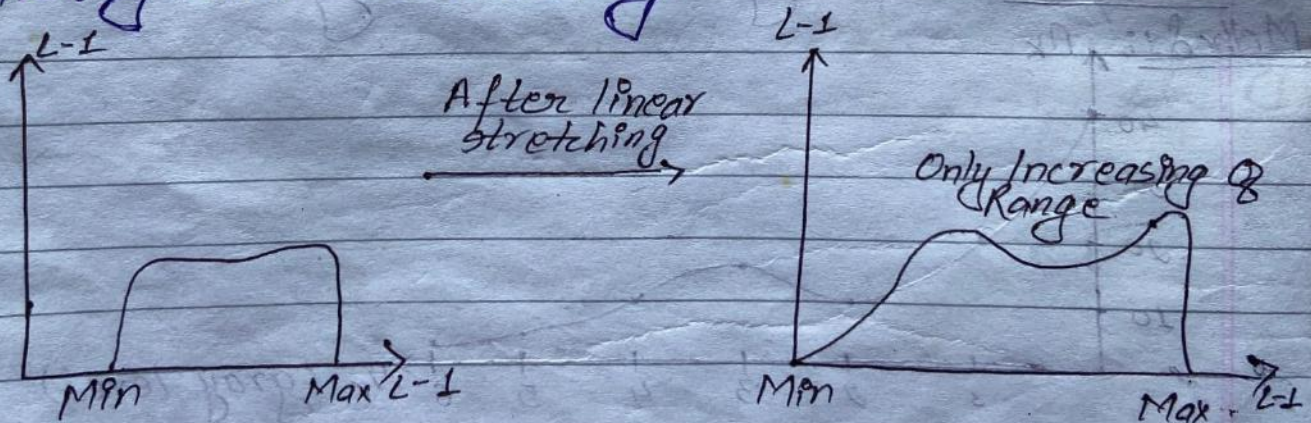


Here,
 $\gamma_k = k^{\text{th}}$ gray level
 $n_k = \text{No. of pixel in } k^{\text{th}} \text{ gray level}$
 $n = \text{total no of pixel}$

Histogram Techniques

- (i) Histogram Stretching / Linear Stretching
- (ii) Histogram Equalization / Histogram Linearization
- (iii) Histogram Specification / Histogram ~~matching~~ ^{matching}

1) Histogram Stretching



(fig: Linear Stretching)

To increase the dynamic range by using a technique known as histogram stretching.

In this method, we don't change basic shape of histogram but spread it over the entire dynamic range. For this we use this technique on the basis of slope. It can be given as:-

$$\text{Slope (m)} = \frac{S_{\max} - S_{\min}}{r_{\max} - r_{\min}} \quad \text{--- (i)}$$

For transformation of histogram stretching then, transformation can be defined as:

$$S = T(r) = \frac{S_{\max} - S_{\min}}{r_{\max} - r_{\min}} (r - r_{\min}) + S_{\min}$$

$$\text{or, } S = \frac{S_{\max} - S_{\min}}{r_{\max} - r_{\min}} (r - r_{\min}) + S_{\min}$$

where,

$S_{\max}$ = max^m gray level in o/p image

$S_{\min}$ = min^m gray level in o/p image

$r_{\max}$ = max^m gray level in i/p image

$r_{\min}$ = min^m gray level in i/p image

This transformation function stretches or increase the dynamic range of given image.

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Q	Gray level	0	1	2	3	4	5	6	7
	No. of pixel	0	0	50	60	50	20	40 ¹⁰	0

Now, perform histogram stretching. so ~~sketch~~ that the new image has a dynamic range of entire range i.e. $[0, 7]$

Solⁿ

$$r_{\min} = 2$$

$$S_{\min} = 0$$

$$r_{\max} = 6$$

$$S_{\max} = 7$$

Now,

we use transformation function of on H.S as:

$$S = T(r) = \frac{S_{\max} - S_{\min}}{r_{\max} - r_{\min}} (r - r_{\min}) + S_{\min}$$

$$S = \frac{7-0}{6-2} (r-2) + 0$$

$$S = \frac{7}{4} (r-2)$$

When $r = 2$, $S = \frac{7}{4} (2-2) = 0$

$r = 3$, $S = \frac{7}{4} (3-2) = \frac{7}{4} \approx 2$

$r = 4$, $S = \frac{7}{4} (4-2) = \frac{7}{2} \approx 4$

$$r=5, \quad S = \frac{7}{4}(5-2) = \frac{21}{4} \approx 5$$

$$r=6, \quad S = \frac{7}{4}(6-2) = 7$$

Hence, modified histogram or stretched range can be illustrated as:

Gray level	0	1	2	3	4	5	6	7
No. of pixels	50	0	60	0	50	20	0	40 10

Now, Output histogram is as shown below:

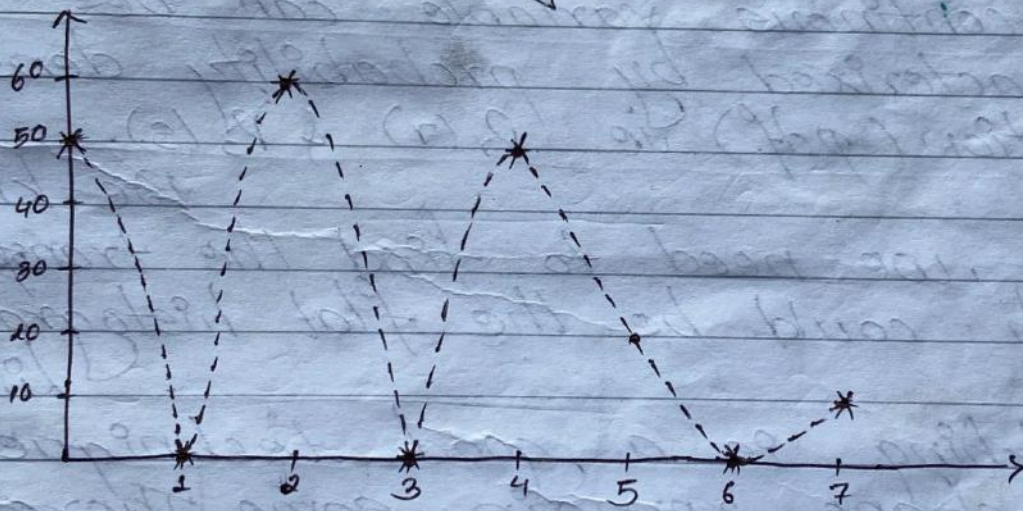
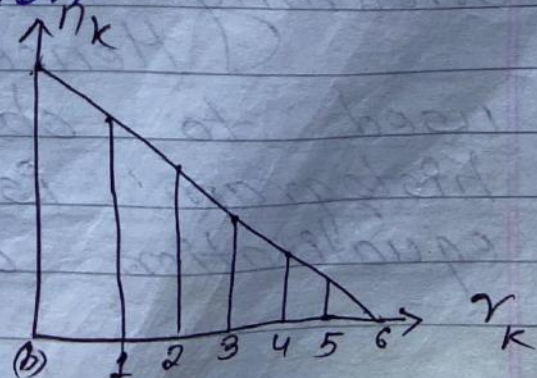
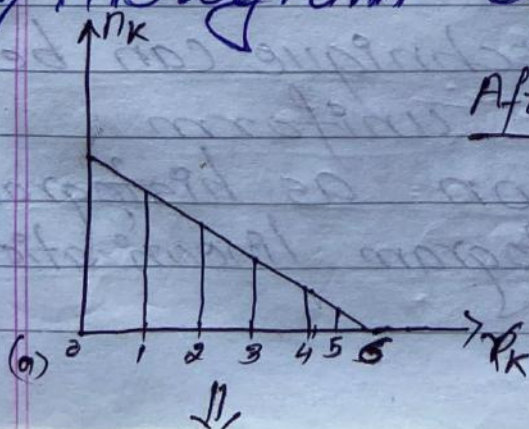
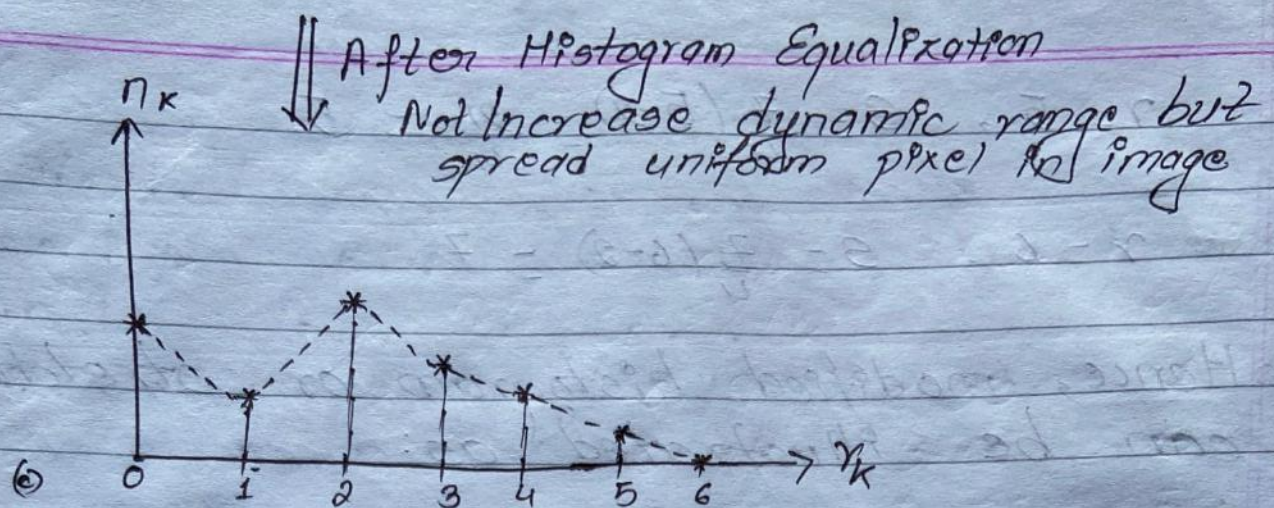


fig: stretched/modified o/p histogram

2) Histogram Equalization





(fig. Histogram Linearization)

The gray level for continuous variable can be characterized by probability density function (pdf) i.e. $P_r(r)$ & $P_s(s)$.

We know that, we need to find the transformation which could be the flat histogram.

Linear stretching is a good technique but shape remain same. There are many application where we need a flat histogram so that this cannot be achieved by histogram stretching.

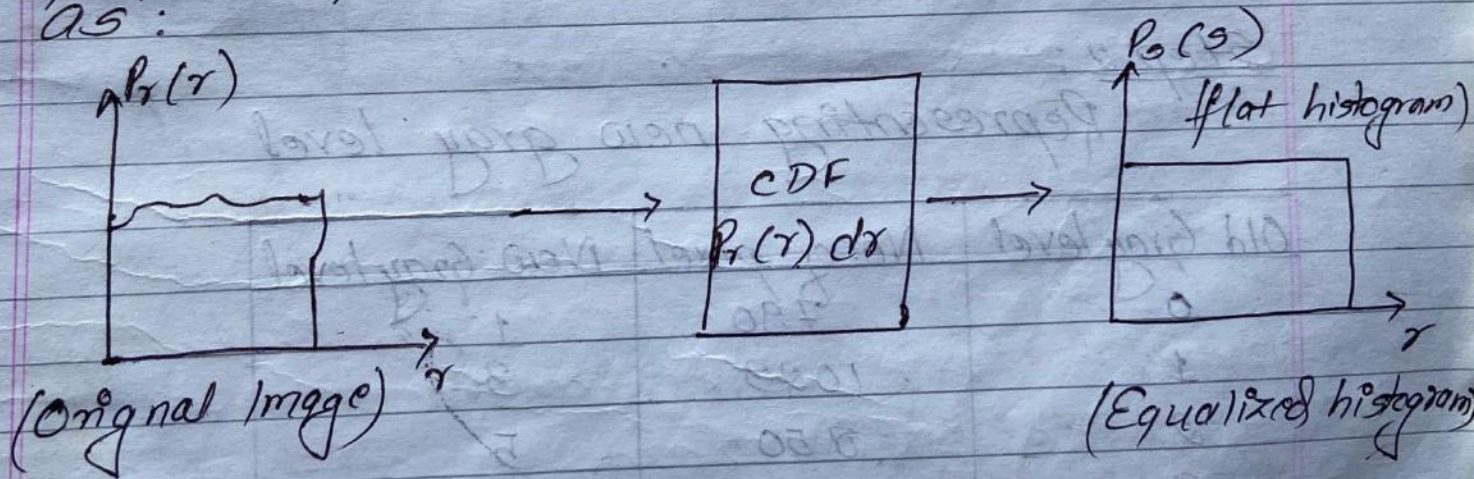
Hence, a technique can be used to obtain a uniform histogram is known as histogram equalization or histogram linearization.

A perfect image is one which has equal number of pixels in all gray level. Therefore, image is not only spread the dynamic range but also to have equal number of pixels in all gray level.

We know that,

$$S = T(r) = \int_0^r P_r(r) \, dr, \quad 0 \leq r \leq 1$$

Above equation shows the transformation function of cumulative pdf (c.d.f) and it can be shown as:



V. Imp ~~Q~~ Equalize the given histogram:

Gray level	0	1	2	3	4	5	6	7
No. of pixel	790	1023	850	656	329	245	122	81

Solⁿ Step 1:

Here,

Gray level	n_k	P.D.F $P_r(r_k) = n_k/n$	C.D.F $S_k = \sum P_r(r_k)$	$7 * S_k$	Rounding off
0	790	0.19	0.19	1.33	1
1	1023	0.25	0.44	3.08	3
2	850	0.21	0.65	4.55	5
3	656	0.16	0.81	5.67	6
4	329	0.08	0.89	6.23	6
5	245	0.06	0.95	6.65	7
6	122	0.03	0.98	6.86	7
7	81	0.02	1.00	7	7
	$n =$ 4096				

Step 2:

Representing new gray level

Old Gray level	No. of pixel	New Gray level
0	790	1
1	1023	3
2	850	5
3	656	6
4	329	6
5	245	7
6	122	7
7	81	7

Step 3:

Modified Histogram

G_n								
Gray level	0	1	2	3	4	5	6	7
No. of pixel	0	790	0	1023	0	850	985	488
								448

Hence, Equalized Histogram:

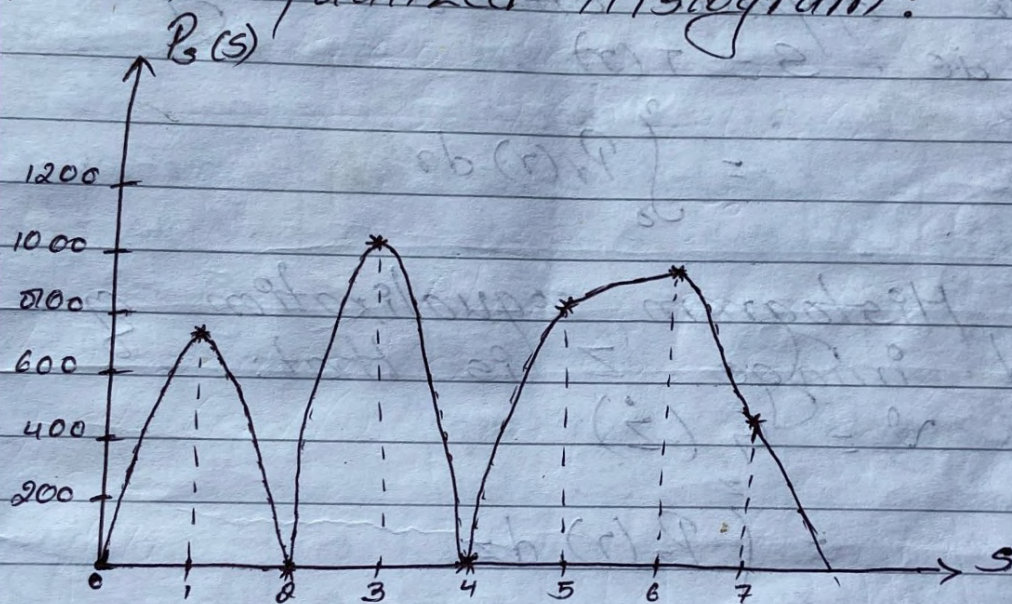


fig: Equalized Histogram

(Th + Nm)

3) Histogram Specification

Histogram matching is not interactive because it always gives one result i.e. An approximation to an uniform histogram.

It is at times desirable to have an interactive method in which certain gray levels are highlighted.

Let us suppose, $P_r(r)$ is the original p.d.f.

$P_z(z)$ is the desired p.d.f

Suppose that histogram equalization is first applied on the original image i.e $S = T(r)$

$$= \int_0^r P_r(r) dr$$

Histogram equalization of desired image 'Z' is that:

i.e $v = G(z)$

$$= \int_0^z P_z(r) dz$$

Now,

The inverse process $z = G^{-1}(s)$ which have the desired p.d.f that provide histogram specification.

i.e $z = G^{-1}(s) = G^{-1}\left(\int_0^r P_r(r) dr\right)$

$$z = G^{-1} \int_0^r P_r(r) dr$$

Q Match the given histogram:

Histogram for image (a)

Gray level	0	1	2	3	4	5	6	7
No. of pixel	790	1023	850	656	329	245	122	81

Histogram for image (b)

Gray level	0	1	2	3	4	5	6	7
No. of pixel	0	0	0	614	819	1230	819	614

Solⁿ Step 1: ~~data~~ ~~data~~

Equalized Histogram of image (a)

Gray level	0	1	2	3	4	5	6	7
No. of pixel	0	790	0	1023	0	850	985	448

Now,

Equalize histogram of image (b)

Gray level	n_k	$P_r(r_k) = \frac{n_k}{n}$	$S_k = \sum P_r(r_k)$	$7 \times S_k$	Rounding off
0	0	0	0	0	0
1	0	0	0	0	0
2	0	0	0	0	0
3	614	0.15	0.15	1.05	1
4	819	0.20	0.35	2.45	2
5	1230	0.30	0.65	4.55	5
6	819	0.2	0.85	5.95	6
7	614	0.15	1	7	7

$$n = 4096$$

Equalized histogram of image (b)

Gray level	0	1	2	3	4	5	6	7
No. of pixels	0	614	819	0	0	1230	819	614

Step 2:

To obtain histogram specification, we apply inverse transformation comparing both equalized histogram.

Gray level	0	1	2	3	4	5	6	7
No. of pixel	0	0	0	790	0	850	985	448

Table (b) $\rightarrow$ Gray level (newly)

Table (a) $\rightarrow$ Pixel (corresponding to new gray level)

Hence, the matched or resultant histogram is as shown below:



(fig: specified image)

Spatial Operations

Many image enhancement techniques are based on spatial operation. Most of the spatial operation are performed on the basis of local neighbourhood of the input pixel.

Some of the example of spatial operation for image enhancement are as follows:

- (i) Neighbourhood Processing
- (ii) Spatial Averaging
- (iii) Smoothing Spatial filter
- (iv) Zooming
- (v) Low Pass, High Pass & Band pass

1) Neighbourhood Processing.

$$\begin{array}{c}
 \begin{array}{c} \text{L} \\ \hline \begin{array}{ccc} -1 & 0 & 1 \end{array} \end{array} \\
 \begin{array}{c} \text{K} \left\{ \begin{array}{l} -1 \left[\begin{array}{ccc} g(x-1, y+1) & g(x, y+1) & g(x+1, y+1) \\ 0 \left[\begin{array}{ccc} g(x-1, y) & g(x, y) & g(x+1, y) \\ 1 \left[\begin{array}{ccc} g(x-1, y-1) & g(x, y-1) & g(x+1, y-1) \end{array} \right. \end{array} \right. \end{array} \right. \end{array}
 \end{array}$$

(fig: 3*3 neighbourhood Pixel)

We change the value of pixel based on the values of its eight (8) neighbours as shown above.

Instead of 3×3 neighbourhood, we can also use 5×5 , 7×7 , etc. Neighbourhood processing deals with the point operation.

The neighbourhood pixel shown above can be masked into window as shown in figure below:

		L		
		-1	0	1
k	-1	w_1	w_2	w_3
	0	w_4	w_5	w_6
	1	w_7	w_8	w_9

(fig: 3×3 window)

The above figure is a one type of template which is also called as mask or window.

To achieve the neighbourhood processing we place this 3×3 mask on the image corresponding to each of the component and get some information at the centre.

and if $g(x, y)$ is the original image then f is the modified image then f is derived as,

$$f(x, y) = g(x-1, y+1) * w_1 + g(x, y+1) * w_2 + g(x+1, y+1) * w_3 + g(x-1, y) * w_4 + g(x, y) * w_5 + g(x+1, y) * w_6 + g(x-1, y-1) * w_7 + g(x, y-1) * w_8 + g(x+1, y-1) * w_9$$

Hence, the sum of the product of the mask coefficients with corresponding pixel take place.

2) Spatial Averaging [Averaging Filtering]

In spatial averaging, each pixel is replaced by the weighted average of its neighbourhood pixel i.e.

$$v(m, n) = \sum_{(k, l) \in W} a(k, l) y(m-k, n-l)$$

where,

$y(m-k, n-l)$ = i/p image

$v(m, n)$ = o/p image

W = chosen window

$a(k, l)$ = filter weight

A common class of spatial average filter has all equal weight - i.e.

i.e.,

$$v(m, n) = \frac{1}{N_w} \sum_{(k, l) \in w} y(m-k, n-l)$$

where,

$$a(k, l) = \frac{1}{N_w}, \quad N_w \text{ is the no. of pixel in window}$$

Now, each pixel is replaced by its average with the average of its nearest pixel value.

Spatial average is used for noise smoothing low pass filtering and sub sampling of the image.

Example of spatial averaging can be represented as:

$$k \begin{cases} -1 & 0 & 1 \\ \begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \end{bmatrix} \end{cases}$$

(fig: 3x3 mask window)

$$k=2 \begin{cases} L=2 \\ \begin{bmatrix} 1/4 & 1/4 \\ 1/4 & 1/4 \end{bmatrix} \end{cases}$$

(fig: 2x2 mask window)

3) Smoothing Spatial Filter

[Median and Mean Filter]

gmv

Median Filter:

The median filter is also a spatial filter but it replaces the central value in the window with median of all the pixel value in the window.

The kernel or template or mask is usually square but can be initiated in different window.

An example of median filtering is:

Unfiltered values in order like
0, 1, 2, 3, 4, 6, 10, 15, 19, 97

Now,

$$\begin{aligned} \text{o/p value} &= \left(\frac{n+1}{2}\right)^{\text{th}} \text{ item} \\ &= 10 \left(\frac{9+1}{2}\right)^{\text{th}} \text{ item} \\ &= 5^{\text{th}} \text{ item} \\ &= 6 \end{aligned}$$

Hence,

6	2	0
3	97	4
19	15	10

(Before filtering)

Median filtering →

6	2	0
3	6	4
19	15	10

(After filtering)

The center value 97 is replaced by the median value of all nine values i.e. 6.

Formula for median filtering is:

$$v(m,n) = \text{Median} \{ y(m-k, n-l) \mid (k,l) \in W \}$$

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Mean filter:

Mean filter is a spatial filter that replace the center value in the window with the average of all the pixel values in the window.

Here, the window can be of any shape but can be initiated as a square mask. Therefore average of smoothing filter shows the properties of low pass filter. So, it reduces the noise from an image.

An example of mean filter is:

Unfiltered values in order like

5, 3, 6, 2, 1, 9, 8, 4, 7

Now,

$$\text{o/p value or mean value} = \frac{\sum x}{N} = 5$$

Hence,

5	3	6
2	(1)	9
8	4	7

(Before filtering)

Mean filtering

5	3	6
2	(5)	9
8	4	7

(After filtering)

The center value '1' is replaced by the mean value '5'.

(N. Imp)

4) Zooming

[Magnification & Interpolation]

Another Important

application is image enhancement where spatial domain neighbourhood operation is used.

Such type of image enhancement technique is known as zooming.

Actually, it is the process of making an image larger. It involves following two steps:

- (i) Creating of new pixel location
- (ii) Assigning of grey level to those new location

Different types of zooming techniques can be used. They are:

- (i) Replication
- (ii) Linear Interpolation

1) Replication

In replication, we simply replicate each pixel and then replicate row.

Let us consider an image as:

1	2	3	4
5	6	7	8
9	8	6	7
0	1	2	3

As start from the first row, first we replicate each pixel and then replicate each row.

Now, The first row looks like:

$[1 \ 1 \ 2 \ 2 \ 3 \ 3 \ 4 \ 4]$

We will now replicate this row then it looks as:

$\begin{bmatrix} 1 & 1 & 2 & 2 & 3 & 3 & 4 & 4 \\ 1 & 1 & 2 & 2 & 3 & 3 & 4 & 4 \end{bmatrix}$

Now, performing replication operation on an entire image then we obtain output image as:

1	1	2	2	3	3	4	4
1	1	2	2	3	3	4	4
5	5	6	6	7	7	8	8
5	5	6	6	7	7	8	8
9	9	8	8	6	6	7	7
9	9	8	8	6	6	7	7
0	0	1	1	2	2	3	3
0	0	1	1	2	2	3	3

Here, 4×4 image is zoomed to 8×8 image.

This method can be repeated to get larger image but no new data is added.

Zooming by zero Interlace Technique

Step 1:

Adding zero to every after pixel of the first row, then we obtain row as:

$[1 \quad 0 \quad 2 \quad 0 \quad 3 \quad 0 \quad 4 \quad 0]$

Step 2:

Adding zero at every column.
Now, insert row of zero & then this is known as zero-interlacing.

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 3 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step 3:

Performing zero-interlace technique on an entire image then we obtain output image as:

$$\begin{bmatrix} 1 & 0 & 2 & 0 & 3 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 5 & 0 & 6 & 0 & 7 & 0 & 8 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 9 & 0 & 8 & 0 & 6 & 0 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 2 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(fig: 8x8 final Pseudo image)

Hence, this image is known as zero-interlace image.

2) Linear Interpolation

In this method, instead of replicating each pixel, average of the two adjacent pixels along the row is taken and placed between two pixels.

Accordingly same operation is then performed along the column.

linear Interpolation is also derived from the zero-interlace technique.

Step 1:

Zero Interlace

1	0	2	0	3	0	4	0
0	0	0	0	0	0	0	0
5	0	6	0	7	0	8	0
0	0	0	0	0	0	0	0
9	0	8	0	6	0	7	0
0	0	0	0	0	0	0	0
0	0	1	0	2	0	3	0
0	0	0	0	0	0	0	0

Step 2

Interpolate row

1	1.5	2	2.5	3	3.5	4	2
0	0	0	0	0	0	0	0
5	5.5	6	6.5	7	7.5	8	4
0	0	0	0	0	0	0	0
8	8.5	8	7	6	6.5	7	3.5
0	0	0	0	0	0	0	0
0	0.5	1	1.5	2	2.5	3	1.5
0	0	0	0	0	0	0	0

Step 3

Interpolate column

1	1.5	2	2.5	3	3.5	4	2
3	3.5	4	4.5	5	5.5	6	3
5	5.5	6	6.5	7	7.5	8	4
7	7	7	6.25	6.5	7	7.5	3.25
9	8.5	8	7	6	6.5	7	3.5
4.5	4.5	4.5	4.5	4	4.5	5	2.5
0	0.5	1	1.5	2	2.5	3	1.5
0	0.25	0.5	0.75	1	1.25	1.5	0.7

gmp 5) Filtering

① Low Pass Filtering

It is employed to remove high frequency (unnecessary noise) from a digital image.

This

type of filtering is usually used to attenuate the image noise.

Here, an image is smoothened by decreasing the disparity between the pixel value by average value.

Example of mask or kernel or window of low pass filter can be shown as below:

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

(fig: 3*3 mask)

$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$
$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$

(fig: 4*4 mask)

(ii) High Pass Filtering

A pass filtering is the basis for most sharpening methods. Sharpening is the opposite of smoothing.

The sharpening can make it possible to highlight the borders between homogenous region.

This is used to return high frequency information.

The kernel of the high pass filter is designed to increase the brightness of center pixel.

Example of mask or kernel or window of high pass filter is as shown below:

$1/9$	$1/9$	$1/9$
$1/9$	$-1/9$	$1/9$
$1/9$	$1/9$	$1/9$

(fig: 3×3 mask high pass filter)

(iii) Band Pass Filter

This filter is used to remove selected frequency region between low and high frequency.

It is also used for image restoration. It passes a limited range of frequency.

Calculating spatial averaging for band pass filtering is:

$$v(m, n; \theta) = \frac{1}{N_0} \sum_{(k, l) \in W_0} y(m-k, n-l)$$

where

W_0 = Neighbourhood mask along directions
 N_0 = No. of pixel within the neighbourhood.